

Answer **all** questions.

1. The table below gives information about four types of power station.

The table ranks the power stations in order from 1 to 4 for three different features.
Rank 1 is best and rank 4 is worst.

Power station	Efficiency	Rank	Running cost	Rank	Emissions	Rank
Type A	25 %	4	Second highest	3	Highest polluting emissions	4
Type B		1	Practically zero	1	No emissions	1
Type C	35 %	3	Highest	4	Has cleaner emissions than type A power stations	2
Type D	40 %	2	Second lowest	2	Cleaner emissions than type C power stations but produces radioactive waste	3

- (a) Use the information in the table to answer the following questions.

- (i) Gareth says that the best type of power station to recommend overall by ranking is **type B**. Explain whether you agree with him. [2]

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- (ii) The energy sources for different types of power station are **fossil fuel**, **nuclear** and **hydroelectric**.

Complete the table below for the energy sources for types **A**, **B**, **C** and **D**. [3]
Each energy source may be used once, more than once, or not at all.

Type	Energy source
A	
B	
C	
D	

- (b) Use the information below and an equation from page 2 to calculate the % efficiency of a **type B** power station. [2]

Input energy = 200 000 MJ
Heat energy produced = 30 000 MJ
Electrical energy produced = 170 000 MJ

% efficiency =

7

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